PROJECT REPORT

Emotion, Sentiment and Sarcasm Analysis of Social Media Text

Natural language processing for short, informal text

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Abstract

This project builds and evaluates a system that reads a short social media post and predicts its emotion and sentiment, and flags possible sarcasm. A TF-IDF and Linear SVM baseline reached 0.963 macro-F1 for emotion and 0.946 for sentiment on a cleaned, balanced dataset. Testing on independent human-labelled text showed that this result did not transfer: emotion accuracy fell to 0.303, because the model had largely learned emotion keywords.

The project therefore moved to transformer models trained on natural comments and tweets. The final system uses one Twitter-based RoBERTa model for emotion and sarcasm, combined with a rule-based emoji vote, and reaches 0.688 macro-F1 across seven classes on a natural test set. It detects sarcasm with F1 of 0.454 on iSarcasm and 0.619 on SemEval irony, compared with 0.339 and 0.540 for a TF-IDF baseline. Sarcasm and emoji meaning remain only partly solved, and results on the hand-built ambiguous test set are provisional because its labels have not yet been reviewed by a person.

1. Introduction

People express emotion on social media through short, informal text that often relies on emojis, slang, and sarcasm. A phrase such as "I just love it when my phone dies at 2%" uses positive words to express annoyance. Models that rely on individual words read such text literally.

Objectives

- Classify the emotion and sentiment of short social media posts.
- Detect and remove label leakage, and measure performance fairly.
- Test whether emoji-aware features improve results on ambiguous text.
- Train a model that handles natural text, sarcasm, and emojis better than the baseline.
- Provide a working Streamlit application for trying the system.

Research question: to what extent can emoji-aware feature extraction improve sentiment and emotion classification on ambiguous social media text, compared with conventional text-only machine learning and rule-based emoji integration?

2. Datasets

Table 1. Datasets used in the project

Dataset	Purpose	Labels	Rows
Project dataset (final.csv)	Baseline training and testing	Emotion: 6 classes. Sentiment: heuristic word list	58,980 after cleaning
GoEmotions (Reddit)	Natural emotion training and testing	Human	37,465
TweetEval sentiment	Independent sentiment testing	Human	12,280 test rows
SemEval-2018 Affect in Tweets	Emotion in tweets	Human annotators	5,521
iSarcasmEval and SemEval-2018 irony	Sarcasm detection	Authors (iSarcasm); annotators (irony)	10,214
Ambiguous social media set	Evaluation only	AI-suggested, pending human review	500

Cleaning the project dataset

The raw data contained 416,809 rows. Exploratory analysis found emoji suffixes that directly encoded the labels, which is target leakage. These were removed so that the model had to use the language in each text. Contradictory texts and exact duplicates were then dropped, leaving 371,527 unique texts.

Emotion and sentiment were balanced at the same time through undersampling, with an integer max-flow step deciding how many rows to take from each emotion and sentiment combination. The final dataset has 9,830 rows per emotion and 19,660 per sentiment. It was split 80/20 into 47,184 training and 11,796 test rows, stratified on both targets.

Sarcasm and tweet data

A separate build script combined the sarcasm sources and removed any text shared between splits or with the ambiguous test set. In the raw irony test file, every ironic tweet contained a hashtag such as #not, and tweets without one were never ironic. These hashtags were removed because they would have revealed the label.

Authors in iSarcasmEval also wrote what they really meant. These 776 plain rephrases were used as non-sarcastic training examples, giving pairs with the same meaning with and without sarcasm.

3. Method

3.1 Baseline models

Text was lowercased, URLs were removed, and whitespace was normalised. TF-IDF features used single words and word pairs, with up to 50,000 features fitted on training data only. A majority-class baseline, Logistic Regression, Linear SVM, and Multinomial Naive Bayes were compared separately for each task.

3.2 Emoji handling

Because leakage removal left no emojis in the baseline training data, the model could not read them. Late fusion was introduced: recognised emojis are removed and counted, the words are scored by the model, and the final score is (1 - w) x text score + w x emoji vote, with w = 0.25 for emotion. A text with no words uses the emoji vote alone.

Four further systems tested emoji features: converting emojis to their names (B), adding separate emoji features (C), and versions of both trained on synthetic emoji data (B', C'). System D is the words-only SVM (A) combined with the emoji vote.

3.3 Transformer models

A fine-tuned DistilRoBERTa model was first trained on GoEmotions with seven classes, adding neutral. The final model uses cardiffnlp/twitter-roberta-base-2022-154m, a RoBERTa model pretrained on tweets, with two outputs that share one encoder: emotion and sarcasm. Each training row updates the output it has a label for. Training used a learning rate of 2e-5, three epochs, batches of 32, a maximum of 64 tokens, seed 42, and square-root inverse-frequency class weights.

3.4 Three versions of the sarcasm model

Table 2. Model versions

Version	Change	Sarcasm threshold
Version 1	Emotion and sarcasm trained together	0.998
Version 2	GoEmotions comments also used as not-sarcastic examples	0.98
Version 3 (final)	Version 2, plus emotion labels for sarcastic tweets taken from their authors' rephrases	0.90

For version 3, each author's plain rephrase was scored by the version 2 emotion model and its emotion assigned to the sarcastic tweet. Only non-neutral rephrases at 60% confidence or higher were kept, giving 462 training tweets, each counted three times. These are machine-generated labels and include some errors, as sampling confirmed.

A sarcasm rule then applies at prediction time. If the sarcasm probability reaches the threshold and the top emotion is joy or love, the best other emotion is used instead, excluding neutral. Sentiment remains with the TF-IDF and Linear SVM model.

3.5 Evaluation design

- Test sets and the ambiguous set were never used for training.
- The best epoch was chosen on validation data, using the mean of emotion macro-F1 and sarcastic-class F1.
- Each sarcasm threshold was the lowest that cost at most 0.005 emotion macro-F1 on validation data.
- Accuracy, macro-F1, precision, recall, and confusion matrices were reported as appropriate.

4. Results

4.1 Baselines on the project dataset

Table 3. Test results on the project dataset (11,796 rows), accuracy / macro-F1

Task	Majority	Logistic Regression	Linear SVM	Naive Bayes
Emotion	0.167 / 0.048	0.957 / 0.956	0.963 / 0.963	0.886 / 0.885
Sentiment	0.333 / 0.167	0.940 / 0.939	0.946 / 0.946	0.877 / 0.877

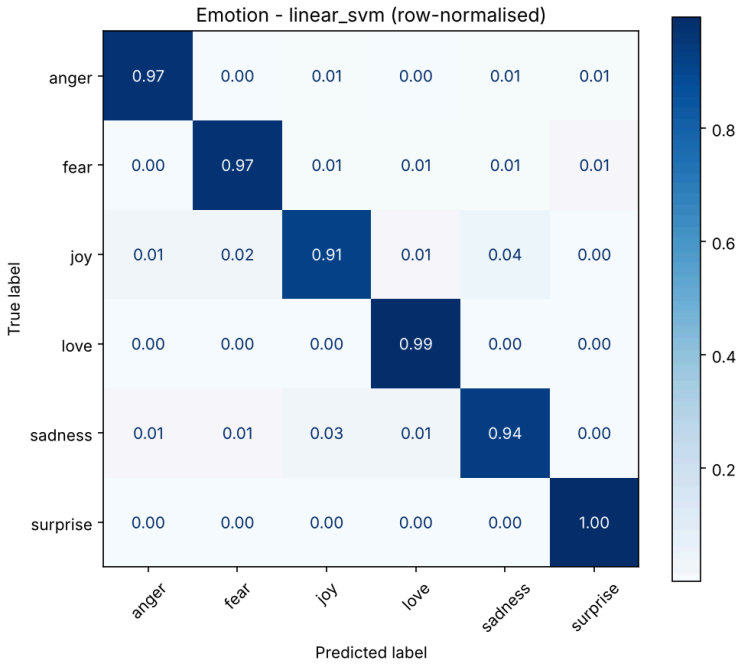


Figure 1. Linear SVM emotion confusion matrix (row-normalised)

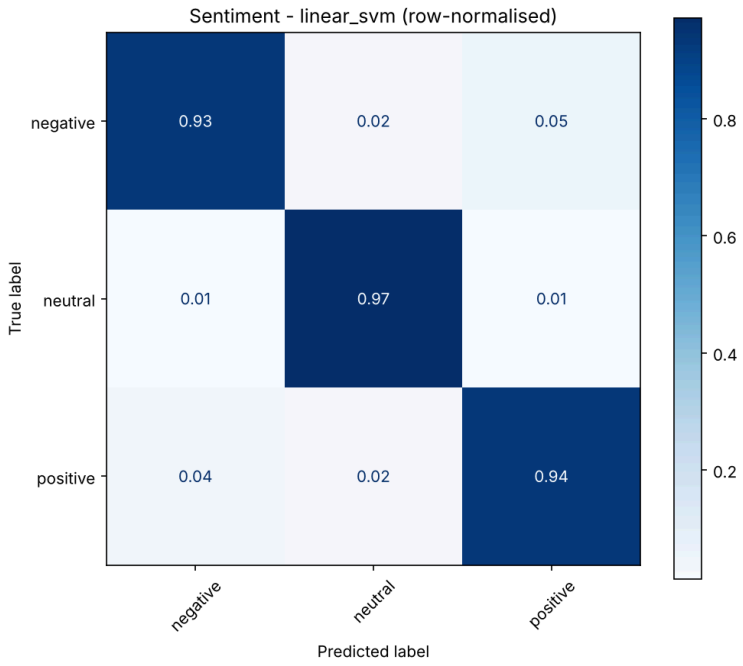


Figure 2. Linear SVM sentiment confusion matrix (row-normalised)

Inspection of feature weights explains these high scores. The strongest emotion features were emotion words themselves, such as "terrified" or "tender", because most training sentences follow an "i feel X" pattern. The sentiment model had learned the word list used to create its labels: in the training data, sadness was labelled positive more often than negative.

4.2 Testing on independent text

Table 4. Linear SVM on human-labelled data

Task and dataset	Accuracy	Macro-F1
Emotion, GoEmotions (six emotions, 2,195 rows)	0.303	0.284
Sentiment, TweetEval (12,280 rows)	0.510	0.426

4.3 Emoji-aware features on the ambiguous set

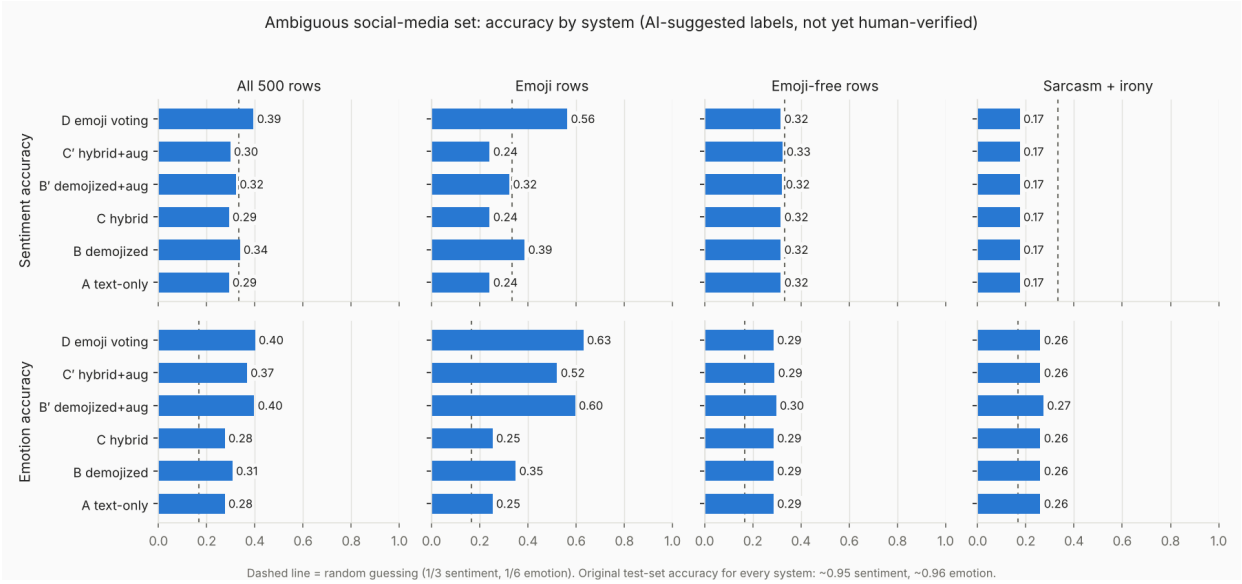


Figure 3. Accuracy by system on the ambiguous set. Labels are AI-suggested and not yet human-verified.

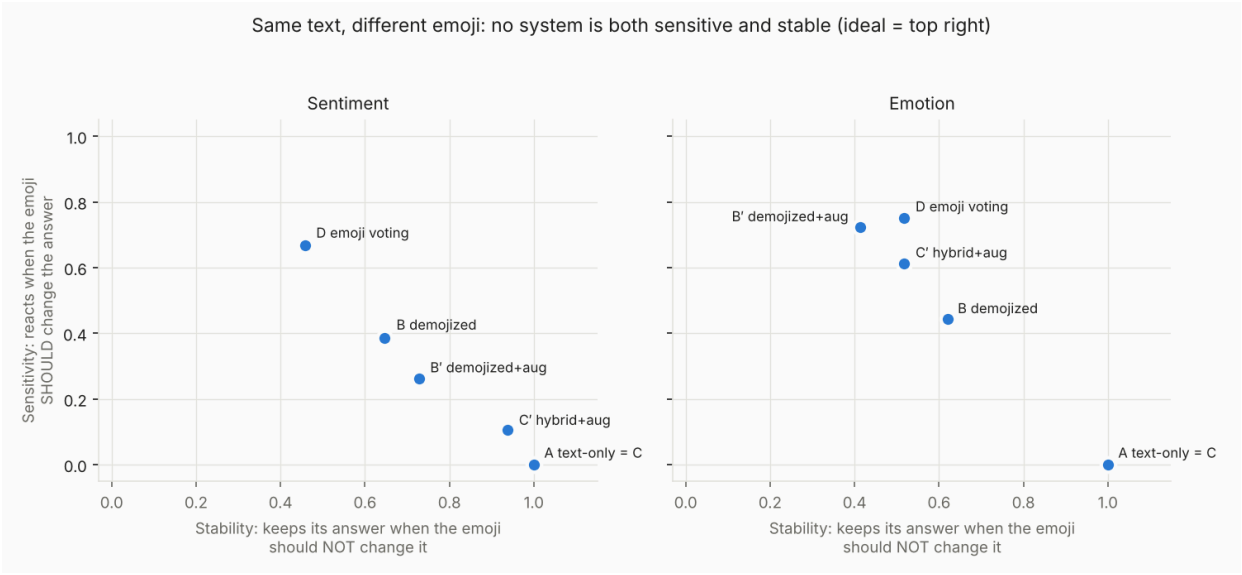


Figure 4. Paired test using the same text with different emojis. No system was both sensitive and stable.

Separate emoji features (C) had no effect, because those columns were always zero in training. Converting emojis to names helped a little. Rule-based voting (D) was the strongest emoji method, but it over-trusted emojis. No emoji method improved sentiment on sarcasm or irony, which stayed at 0.17.

4.4 Transformer and sarcasm models

Table 5. Comparison of systems at their validation-chosen settings

Measure	System D (SVM + emoji)	DistilRoBERTa + emoji	Version 1	Version 2	Version 3 (final)
Natural test macro-F1	not comparable	0.670	0.684	0.695	0.688
Ambiguous set, overall	40.1%	35.3%	41.3%	43.6%	42.4%
Sarcasm rows (n=40)	22.5%	2.5%	32.5%	20.0%	27.5%
Positive words, negative meaning (n=40)	35.0%	5.0%	15.0%	17.5%	32.5%
All emoji rows (n=146)	63.0%	51.4%	54.8%	54.1%	50.0%

Ambiguous-set figures hide the neutral class, because those labels never use it. The SVM cannot be compared on the seven-class natural test because it has no neutral class. On the six emotions alone, its accuracy was 0.303, against 0.840 for version 1 and 0.834 for version 3 with no sarcasm rule.

Table 6. Sarcasm detection on held-out tweets at each model's threshold (precision / recall / F1)

Test set	TF-IDF baseline F1	Version 1	Version 2	Version 3 (final)
iSarcasm (1,394 rows)	0.339	0.855 / 0.235 / 0.369	0.848 / 0.195 / 0.317	0.705 / 0.335 / 0.454
SemEval irony (784 rows)	0.540	0.986 / 0.222 / 0.362	0.964 / 0.260 / 0.410	0.783 / 0.511 / 0.619

Version 3 was selected because sarcasm was the main goal and its F1 was highest on both larger sarcasm test sets. Without any sarcasm rule, its accuracy on sarcasm rows was 25%, compared with 0% for version 1. This shows that the emotion output itself learned from the rephrase-labelled tweets. Version 3 missed two pre-set targets: sarcasm rows (11 of 40, against 13 of 40 for version 1) and emoji rows.

5. Application

A Streamlit application accepts a post and shows the predicted emotion and sentiment, score charts, the emojis recognised, and a note when sarcasm is detected. It warns on empty input and when no words are recognised. The application was checked with automated runs on a range of inputs, including emoji-only, blank, and nonsense text.

Known errors include "just finished my exams" followed by a party emoji being predicted as sadness, and "Oh great, another meeting that could have been an email" being predicted as love.

6. Discussion

Key findings

- High scores on one dataset can hide a model's dependence on that dataset's vocabulary. Separately collected test data revealed this.
- Removing leakage is essential, and it appeared twice in this project: as label-coded emoji suffixes and as irony hashtags.
- Emoji-aware features help where emojis carry the meaning, but do not solve ambiguity in general.
- Training the emotion output on sarcastic text labelled through author rephrases improved how it reads sarcasm.

An error found during integration

At a threshold of 0.5, version 1 appeared strong on the ambiguous set, with 45.0% overall and 50% on sarcasm rows. Running the whole pipeline on ordinary text showed natural macro-F1 falling from 0.687 to 0.569, and "I love my mom so much" being predicted as anger. Its sarcasm output had only seen tweets, and flagged about a quarter of ordinary comments. This led to the validation-based threshold and to training on ordinary comments as negative examples. It showed the importance of evaluating the exact setup that will be used.

7. Limitations

- All 500 ambiguous labels are AI-suggested and have not been reviewed by a person. The same AI tool wrote the emoji table and the synthetic emoji pools, which may favour systems using them.
- Some ambiguous subsets are small. With 40 sarcasm rows, one row changes the score by 2.5 points.
- Version 3 misses about two-thirds of sarcasm on iSarcasm, and about 3 in 10 of its flags there are wrong.
- Emoji handling is below the older rule-based system, with 50.0% against 63.0% on emoji rows.
- Sentiment still comes from the baseline SVM, which was trained on heuristic labels and ignores sarcasm.
- Sarcastic training labels are machine-generated. Irony is used as a stand-in for sarcasm in SemEval rows.
- Love (21 training rows) and surprise (50) are scarce in the tweet emotion data.
- Baselines used a single 80/20 split, without cross-validation. The system handles English only.

8. Future Work

- Have the 500 ambiguous labels reviewed by two people and measure their agreement.
- Improve emoji understanding with more real emoji-labelled data.
- Train a Twitter-based sentiment model that takes sarcasm into account.

- Add conversation context, since some sarcasm cannot be resolved from one post.
- Use cross-validation and deploy the application online.

9. Conclusion

The project moved from a high-scoring keyword classifier to a system evaluated on natural text. The final model improves emotion recognition on independent data and detects sarcasm better than the TF-IDF baseline. Its remaining weaknesses are clear: most sarcasm is still missed, emoji meaning is only partly handled, and sentiment needs replacing. The evaluation work, particularly finding leakage and testing the complete pipeline, was as important as the choice of model.

AI Assistance Statement

AI tools were used in this project. Claude, by Anthropic, assisted with writing and debugging code, running experiments, suggesting the labels for the ambiguous test set, and drafting documentation and this report. Decisions on the project's direction and on which final model to use were made by the project author.

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